Kongunadu College of Engineering and Technology
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Accredited by NBA (CSE, ECE, EEE, Mech), NAAC,
Recognized by UGC with 2(f) & 12(B)An ISO 9001:2015 Certified Institution
Namakkal-Trichy Main Road,
Thoulrpatti(Post), Thottiam, Trichy-621215, Tamil Nadu

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Academic Year- 2025-26

Internship Report

Name of the Student	DHARSHINI V
Register Number	621324106015
Year/Semester	II-A / III
Email ID	dharshinivelu47@gmail.com
Name of the Industry	UNITED SOFT TECH, SALEM
Internship period (From - To)	24.11.2025 to 10.12.2025
Academic Mentor	**Industry Mentor**
Dr. B. UMARANI **Professor / ECE**	**Mr.V.KAVIYARASAN** **Embedded System Engineer, Salem**

ANNA UNIVERSITY: CHENNAI

DECEMBER 2025

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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PROGRAM OUTCOMES

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES

PSO1: Students shall have skills and knowledge to work and design on PCB, analog and digital systems, adhoc and sensor networks, embedded and communication systems.

PSO2: Students are able to perform and design in the simulation tools and IoT related modules.

BONAFIDE CERTIFICATE

This is to certify that the internship entitled **"HANDS ON INSTRUCTION FOR ADVANCED CONTROLLER TRAINING BASED ON EMBEDDED SYSTEMS"** has been carried out successfully by **DHARSHINI V** and Reg. No. **621324106015** for the completion of II Year III semester Bachelor of Engineering in Electronics and Communication Engineering.

Academic Mentor **HOD**

INTERNSHIP TRAINING REPORT

1. About the Industry:

United Soft Tech, a known firm in Salem for projects, offers embedded systems internships focusing on real-world skills like microcontroller the programming (Arduino, PIC), sensor integration, IoT protocols, embedded C/Python, and hardware-software interfacing, leading to practical experience in building prototypes and solving industry problems, excellent for entry-level careers. Interns typically build IoT solutions, learn data acquisition, and develop project management skills, aiming to bridge theory with practical application in areas like smart devices. Through its training and internship programs, UNITED SOFT TECH helps students enhance their technical knowledge, problem-solving abilities, and professional skills, preparing them for successful careers in the engineering and technology fields.

2. Description of the Internship:

During my internship at United Soft Tech, I received comprehensive hands-on training in advanced embedded systems controller design, which strengthened my understanding of hardware and software integration in real-world applications. The internship focused on embedded programming using C and C++, microcontroller interfacing, and firmware development for platforms such as ARM Cortex–based controllers and Raspberry Pi. I gained practical exposure to real-time operating system concepts, including task scheduling, synchronization, and real-time performance management, enabling me to design reliable and time-critical embedded systems. I also worked on IoT, automation, and robotics-based projects, where I implemented sensor interfacing, actuator control, and communication protocols such as UART, I2C, and SPI. Emphasis was placed on power-efficient design, debugging, and system optimization, which are essential for modern embedded products. This internship effectively bridged the gap between theoretical knowledge and industrial practices, helped me build a strong project portfolio, and significantly enhanced my employability for embedded systems engineer and firmware developer roles, providing a solid foundation for future work in the embedded and electronics industry.

3. Outcomes of the Internship:

- Gained hands-on experience in embedded system development using microcontrollers, sensors, and peripheral devices with industry-standard tools.
- Learned the complete process of embedded product development, from circuit design and firmware coding to hardware integration and testing.
- Understood how client-specific requirements are converted into functional embedded system solutions.
- Enhanced problem-solving skills, debugging ability, and accuracy in programming, interfacing, and handling electronic components.
- Improved teamwork, communication skills, time management, and the practical application of academic knowledge in a professional embedded systems environment.

PO & PSO MAPPING

POs & PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
Internship Outcome	3	2	1	-	2	-	-	2	2	-	2	3	2

Note: 1 – Low level, 2 – Medium level, 3 – High level

Academic Mentor **HOD**

ANNEXURE - I

REQUISITION LETTER

ANNEXURE - II

APPROVAL LETTER

ANNEXURE – III
INTERNSHIP CERTIFICATE

ANNEXURE – V

STUDENT FEEDBACK FORM

ANNEXURE – VI

INTERNSHIP EVALUATION REPORT